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Solar grade silicon: Technology status and industrial trends

Gøran Bye [a,*], Bruno Ceccaroli [b]

[a] Advanced Materials Management Solutions, LLC, 1049 S Laguna Drive, Moses Lake, WA 98837, USA
[b] Marche AS, PO Box 8309, Vaagsbygd, N-4676 Kristiansand, Norway

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ABSTRACT

Crystalline silicon remains (all variants included) the dominant technology to manufacture solar cells. Currently (2012–2013) more than 90% of all solar cells produced are based on this vast group of technologies. The availability, the cost and the quality to the silicon feedstock is therefore a strategic issue of paramount importance for the entire photovoltaic sector. The silicon demand/supply balance has evolved from a situation of shortage with rocketing sales prices, in the years 2005–2008, to currently (2012–2013) an oversupply situation with record low price level for virgin polysilicon. Between these two extreme periods, production capacity has been multiplied by a factor of nearly 10. A better understanding of the prevailing dynamics in the polysilicon/silicon industry is needed in order for all players in the solar cell industry to make proper planning. In light of the past developments as well as the constraints imposed by a sound competition, the present article reviews the market trends for solar grade silicon including capacity, supply, demand and price. Furthermore, the article reviews the competing commercial technologies i.e. Siemens polysilicon, fluidized bed reactor/FBR polysilicon and upgraded metallurgical/UMG silicon and compares them in terms of maturity, improvement potential, product morphology, purity, applications and cost (actual vs. potential).

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1. Introduction

Driven by the dynamics of the PV-market, the industry for manufacturing high purity silicon suitable for solar cells has gone through a dramatic development during the last decade evolving from under- to over-supply, from sky high to dumping price, from insane profits to negative margins. During this period (2003–2013) the global solar grade silicon capacity has been multiplied at least by a factor of 10. In the middle of the last decade hundreds of projects were announced to expand production capacity (both through debottlenecking, brown field projects and green field projects) as well as to develop new low energy, low cost processes more suitable for solar cells than the traditional and proprietary high cost, hyper purity polysilicon processes. A decade ago the industry was dominated by a group of around 10 companies (the legacy producers/incumbents) all located in the EU (Germany and Italy), the USA and Japan. Now the industry may be analyzed as composed of three groups of producers:

- A first group is formed by the four majors; among them two legacy producers and two Asiatic new entrants. The strategic intent of each of these four majors is to target a market share of at least one quarter of the global market, seemingly leaving no room for others—however, lately their expansion plans have been delayed due to the current market conditions.
- A second group is formed by four to six second tier manufacturers, two of which are legacy survivors. Their cumulative capacity may equal half the total demand in 2012.
- A third group is composed of the remaining incumbents and of new entrants, all located in Asia (China, Taiwan, Korea) and Middle East (Saudi Arabia, Gulf emirates) whose future is uncertain.

To be complete one should mention the group of all tens to hundreds projects which never got launched, or which failed and for some of them ended up in bankruptcy.

During the tumultuous past 10 years, where the polysilicon industry, like the PV-industry, initially scrambled for establishing increased capacity at nearly any cost, technology developments as well as economies of scale have nevertheless resulted in lower energy consumption, operating expenses and capital expenditure for solar grade silicon contributing to the overall strong cost reduction required to make solar cells competitive in the energy landscape. New processes have been developed, the most notable being the use of fluidized bed reactors/FBR in polysilicon deposition and the replacement of chemical purification (e.g. distillation) by metallurgical refining. However, the vast majority of solar grade

* Corresponding author. Tel.: +1 509 760 9265.
E-mail addresses: goran.bye@advancedmaterialsmanagement.com (G. Bye), bruno.ceccaroli@online.no (B. Ceccaroli).

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silicon (> 90%) is still produced by the historical so called "Siemens" process applying chemical vapor deposition/CVD of high purity trichlorosilane/TCS/$SiHCl_3$ on a hot filament as this class of process currently is the only one available from technology suppliers and engineering firms. Less than 10% of all solar grade silicon is currently manufactured through the monosilane/SiH_4 deposition process in FBR in spite of its lower cost. The reason for this limited share is to be found in the high technology entry barrier as this process is, for the time being, highly proprietary and well protected by two successful producers. Solar grade silicon through the metallurgical route is promising from a cost point of view. However, for purity reasons it is still struggling to be accepted by the solar cell makers and the long list of these projects emerging during the last decade is now reduced to only a couple of companies remaining in the race.

This article reviews the technical status of all process categories aiming at manufacturing solar grade silicon with emphasis on the CVD ("Siemens") and FBR categories. Cost (of capital as well as of operations), energy consumption, product quality and product adequacy will be compared for each of them. The paper will also point out the challenges and the promises of solar grade silicon in line with the future growth of the photovoltaic industry.

"Polysilicon" is a commonly used term which we will use in this article to refer to any chemical purification process and product going through synthesis and purification of a silicon bearing volatile compound and its decomposition to elementary silicon for the purpose of making semiconductors or solar cells. We also use the generic designation "UMG" in the present article for any silicon purification process and product achieved by metallurgical routes. "Solar grade silicon" refers to any grade of silicon usable in manufacturing solar cells, including polysilicon and UMG. "Semiconductor grade silicon" refers to the higher purity grades of polysilicon usable in manufacturing semiconductors.

2. Production capacity, supply and demand, price development

2.1. A ten year rollercoaster ride

Few predicted 10 years ago that polysilicon, a peculiar metalloid then used as feedstock by a handful of producers of silicon wafers for semiconductors, soon should become one of the hottest topics discussed by investors and analysts worldwide as well as the subject of thousands more, but largely less insightful analyses, articles and reports—due to the attractiveness of the new large application area for polysilicon: photovoltaics/PV.

As governments poured subsidies into PV in the early 2000s, investors followed suit tempted by the significant profits to be had. PV grew extremely rapidly in both volumes and revenues and soon the small number of somewhat staid and definitely unglamorous chemical companies making polysilicon found their capacities overbooked and the sales price of polysilicon skyrocketing.

Looking for seemingly easy profits as well as (in the case of PV-companies) the daunting need to overcome difficulties obtaining this necessary raw material, PV-players, chemical companies, utility and industrial companies – from textiles to fuel cells – and sundry other entrepreneurs first announced and then commenced projects for establishing new polysilicon production. Some succeeded – in a couple of instances spectacularly so – but most failed.

Nevertheless, the capacities from the successful new entrants and the expansions of the incumbents together turned the supply and demand balance from heavily undersupplied in 2005–2008 to well oversupplied in 2011 (and to date). Coinciding with a similar, and maybe even stronger, oversupply situation in other parts of the value chain, polysilicon prices have plummeted to levels below the previous slump in 1998–2003. Both PV- and polysilicon companies find themselves with overextended balance sheets from the strong investment activity in new production capacity and market channels, coupled with underutilized capacity for which debt has to be serviced even though the assets may have been written down or entirely off.

This begs the question: What now?

2.2. Continued demand growth is dependent on continued price reductions

The last time the polysilicon industry found itself in the same situation, albeit on a much smaller scale, it was saved by the emergence of PV as the new dominant application segment with strong volume growth. Electronics, the other application segment, displays a paltry single digit percentage growth and today constitutes 10% or less of the polysilicon demand. No new large application is on the horizon and the polysilicon industry remains dependent on PV for its future growth.

During what many would call points of crisis for the industry in 2009 and again in 2012-2013, PV volumes have continued to grow strongly, with an increasingly wider geographical distribution and acceptance. Since 2010, global PV-installations almost doubled to about 30 GWp in 2012. In 2010 German PV-installations made up roughly half the European share of around 80% of the global installations. In 2012, the European share of worldwide installations fell under 50% – Germany still constituted half of that – and this trend is confirmed by preliminary 2013 numbers.

As always there are several factors impacting the behavior of market players and thus the development of any industry. The rapid reductions of European subsidy schemes and increases in those of other countries – especially China and Japan – could explain the geographical shift, but this is not a satisfactory explanation of the absolute growth. The main reason must therefore be that the PV-growth is driven by the attractive financial returns to PV-system investors relative to alternative investments of similar risk. The returns became attractive as PV-prices were reduced.

We draw the conclusion that PV-installations compete in a market that is price-elastic, that the market is far from saturated, and that investors in PV-systems care for financial returns.

While the above conclusions are very reassuring for the industry as a whole, they are hardly heartening for individual companies that suffer severe losses at the established price levels, especially if the thesis that price reductions will have to continue is correct. Undoubtedly, the price decrease particularly from 2009 to 2013 is dramatic, but is in fact bringing sales prices back in line with the historic price experience curve – a relationship that was temporarily obfuscated by the introduction of the German subsidies – see Fig. 1. The price experience curve shows the development in PV-module sales price measured in USD per Wp as cumulative installed base doubles.

Objections can be raised that the experience curve is no law of nature and the financial return of a PV-system is made up of many more aspects than the price of the PV-modules:

- Total installed cost of system: module price plus balance of system cost (land, inverter, trackers, brackets, cabling, etc.) and installation cost.
- Insolation levels or hours of usable sunlight: all locales in the world do not get the same hours of sunlight per year; the quality of the sunlight (angle of incidence; cloud cover; haze; temperature) impact the power generation that may vary tremendously even with identical PV-modules installed.
- Cost of capital/required rate of return: interest rates directly impact the bottom line.

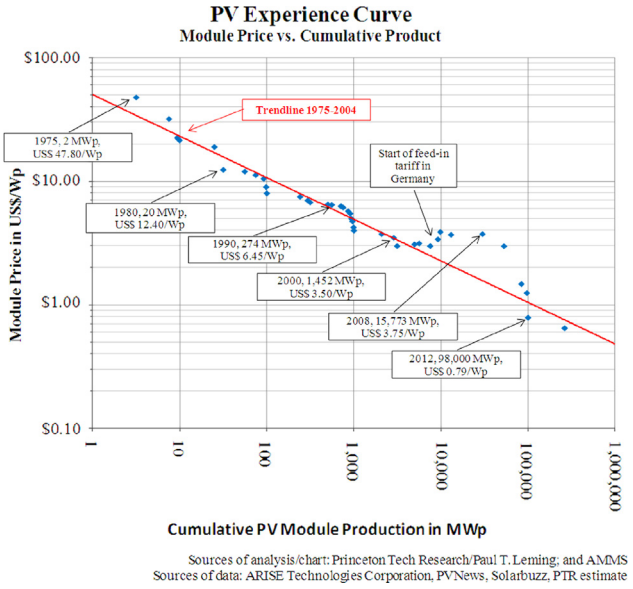


Fig. 1. The PV-module price experience curve.

- Assumed useful life of system: 20 years, 25 years, etc.
- Yield loss over time: Light induced degradation/LID; varying kW h/Wp between PV-modules from different producers in spite of identical Wp-rating.
- Tax regimes (depreciation policies)

There are also very good reasons to measure cost per kWh electricity generated – Levelized Cost of Energy/LCOE – particularly when comparing PV to other power sources. (See for instance [1] for more on LCOE.)

Others, to whom the price developments indicate chilling prospects, argue that at current price levels PV-companies cannot survive. For polysilicon's part, arguments are that polysilicon cost no longer has the same prominence in the PV-cost stack as it had five years ago, and that focus should be on solar conversion efficiency and balance of system costs. Increasing demands for higher sunlight conversion efficiency and guaranteed 25 year lifetime may justify a higher priced high purity polysilicon because of savings elsewhere in the cost stack.

Without question, the efficiency of the PV-modules matters as a cost dilutor as it is the denominator of the total installed cost fraction. Quality also matters as it impacts both the lifetime and degradation of efficiency over time. Likewise, indications are that the prices of the balance of system components and labor hitherto have not followed the same steep reductions as the price for PV-modules. Finally, compared to conventional power plants, the cost of PV-generated electricity is purely an amortized/calculated cost as the fuel is free, operations and maintenance cost nearly negligible and the cost is essentially limited to the initial capital expense—much like, in particular, wind and nuclear power.

We do not want to wade into these discussions. We do, however, firmly believe that the PV-industry will innovate and develop itself into a position where sound companies will not only survive, but thrive under the future price regimes once the ongoing attrition and consolidation has played out in the industry.

Our objective is to form an opinion on the current status and future development of polysilicon technologies and to do this we need to derive some sort of guidance on the future price level for polysilicon on the following premises:

- Polysilicon demand is dependent on growth in PV-volumes. For PV to continue to grow, the decreasing trend in average sales prices for PV-systems has to continue. Even if the relative price of PV-modules lately have shrunk from around 50% to 40% of total PV-system price, PV-modules have to continue their path down the price experience curve to defend against reduced subsidies and to prepare for the potentiality of increasing interest rates (ignoring a problem seldom makes it go away). Prices and costs for PV-modules and their components are best normalized and operationalized on a per Wp-basis.
- Further, we are of the opinion that price determination for polysilicon is made top-down from the market price of PV and not any longer bottom-up based on the cost of polysilicon production.
- Additionally, we think that each link in the value chain – including polysilicon – must be viewed as individual profit generating businesses where the market determines price and where cost is covered and return on assets generated based on the market prices even if a particular polysilicon business should happen to be integrated with other businesses and actual intra-company transfer prices set following other principles than the market price.
- Finally, loss making operations are not sustainable in the long run.

2.3. Longer term polysilicon price to PV cannot exceed USD 20 per kilogram

Based on the above and applying reasonable expectations for improvements in specific silicon usage per Wp as well as reasonable expectations for the necessary size of a gross margin pool for the parts of the value chain making ingots, wafers, cells, and modules – these businesses are after all not very capital intensive, and do portray some features of commodities – it is difficult to argue for a longer term polysilicon price that exceeds USD 20 per kg.

This is depicted in Fig. 2 where it is shown that to believe in a polysilicon price of USD 20 per kg (or more), one must believe that specific usage of silicon can be lowered to 4.5 g per Wp (or below) and that cost of converting the polysilicon to finished PV-modules can be USD 0.405 per Wp (or less) if the PV-mid stream – ingots, wafers, cells, and modules – shall be allowed 10% gross margin – USD

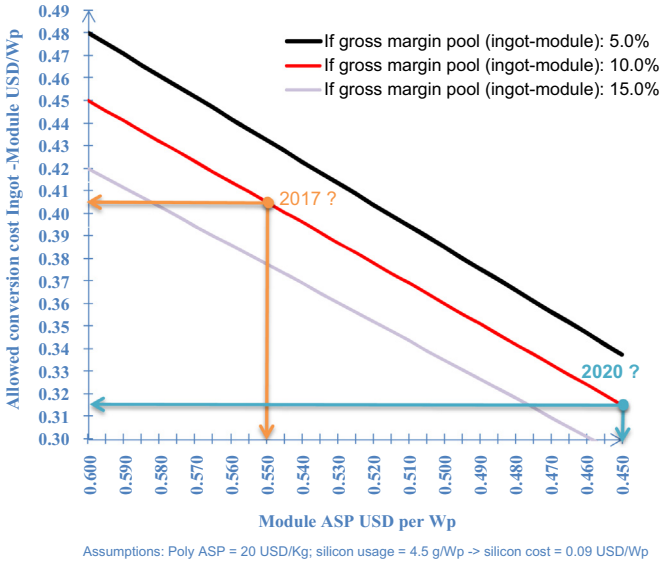


Fig. 2. "Allowed" conversion cost from polysilicon to PV-module at various assumptions on sales price and desired gross margin given polysilicon price=USD 20 per kg, and specific usage=4.5 g of polysilicon per Wp at the module level.

0.055 per Wp – on a module sales price of USD 0.55 per Wp. If the mid-stream gross margin pool must be 15% to cover Sales-, General- and Administrative cost/SG&A, finance cost etc and leave a net profit, the allowed conversion cost is reduced to USD 0.38 per Wp; if 5% gross margin is sufficient, the conversion cost can be USD 0.43 per Wp.

Extrapolating the PV-experience curve indicates that PV-module sales price of USD 0.55 per Wp may occur in 2017 and USD 0.45 per Wp in 2020.

Polysilicon may not be the most prominent layer of the PV-cost stack, but it is clearly not irrelevant as price decreases continue and cost has to follow.

The term "longer term price" is used carefully because first there is no way to predict precisely when also this price level becomes unsustainable, but based on the premises for the discussions in this article, we are of the opinion that this "longer term" will cover most of the time until 2020; and second, that there are several developments that could push the current polysilicon price temporarily off this path—one of these forcing functions is the development in the supply and demand balance.

2.4. Supply and demand balance until 2020—Significant additional capacity needed from 2017

Trying to estimate the world's total polysilicon capacity is like trying accurately to measure a rubber band—the sheer number of project announcements marred by low precision in scale and timing and varying levels of veracity makes it hard to differentiate between what is real and what is not. Based on our knowledge of the industry and applying our best judgment on company announcements and reports, we estimate the relevant polysilicon production capacity for PV to be around 250 000 Metric Ton per year (MT/y) at the end of 2012 with another ~80 000 MT/y underway in the shape of expansion projects, some of which are delayed awaiting market developments, but likely to be in operation by the end of 2015. Expressed in annual PV-module production capacity, there should be sufficient polysilicon for around 45 GWp exiting 2012 and 60 GWp at the end of 2015—see Fig. 3. The existing capacity is predominantly Siemens chemical vapor deposition/CVD-reactors using trichlorosilane/TCS as feed gas. Around 10% is fluidized bed reactors/FBR using monosilane as feed gas.

Plotting these estimates against a forecast for PV-installations until 2020 (amalgamated from several prognoses and discussions), a preliminary supply/demand balance may be achieved already in 2014. Should this balance materialize, it will trigger temporary price spikes as well as completion of halted expansion projects in addition to the other significant ongoing polysilicon projects that can be expected to go live by 2015, potentially creating another short period of oversupply.

Please note that if the numbers are close to being correct, significant net additional polysilicon capacity is needed in the second part of the current decade.

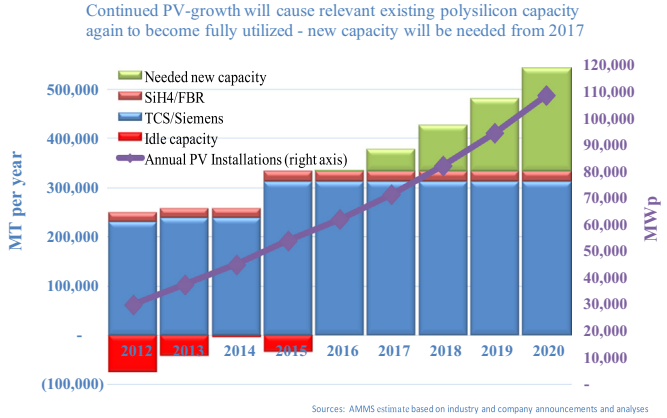


Fig. 3. Polysilicon supply and demand balance towards 2020, polysilicon capacity per technology.

2.5. Not all polysilicon capacity is created equal

From a price determination perspective the above development in the supply and demand balance poses a conflict. In principle, sales prices should increase when capacity is tight to entice idle capacity to be put back in operations and new capacity to be built. Also in principle, it is less risky and more economical or a producer to increase sales prices than to expand capacity.

On the other hand, sales prices cannot increase substantially if the growth in PV and consequently polysilicon demand shall rise to the levels indicated in Fig. 3. We are of the opinion that the market is price-elastic and will set the sales price top-down. We do not see any drivers that would significantly raise the trend in PV-sales prices in the period we are reviewing. On the contrary, potentially rising interest rates will put the PV-sales prices under more pressure as PV will have to become truly competitive to other power generation technologies without being able to enjoy the high levels of subsidies of the past.

Please bear in mind when evaluating price development versus our postulate of longer term polysilicon sales price not exceeding USD 20 per kg, that average spot prices since third quarter 2012 have been below this level and that for the entire first half of 2013 the export price from the USA to China was well below USD 15 per kg, so far bottoming out at USD 12 per kg on substantial volumes according to Chinese customs statistics.

This leads to a different perspective on the existing polysilicon capacity. Building on a widely used overview from Bloomberg New Finance/BNEF [2], Fig. 4 shows that only about 90 000 MT/y of existing capacity is reported to have "variable cost of production" of USD 18 per kg or below. We will come back to cost definitions later, but will simply accept the BNEF estimates for now. At a sales price of USD 20 per kg these 90 000 MT/y will yield USD 2 per kg to cover SG&A, finance cost, profit etc. All the remaining capacity will generate losses at this sales price.

Loss making operations are not sustainable in the long run and thus price escalations could be expected as owners of the capacity above the lowest cost 90 000 MT/y search profits. But: as most polysilicon capacity was built over the last 10 years, there are a large number of relatively new production plants that have been partially or totally written off by their owners as sunk cost. Balance sheets have been restructured and debt refinanced. Further, several

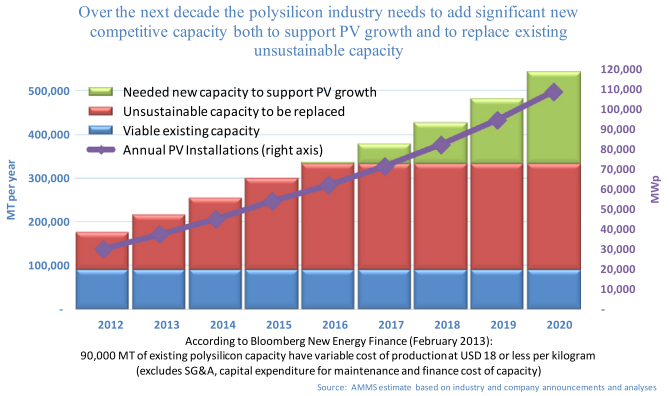


Fig. 4. Polysilicon supply and demand balance towards 2020, polysilicon capacity per cost level.

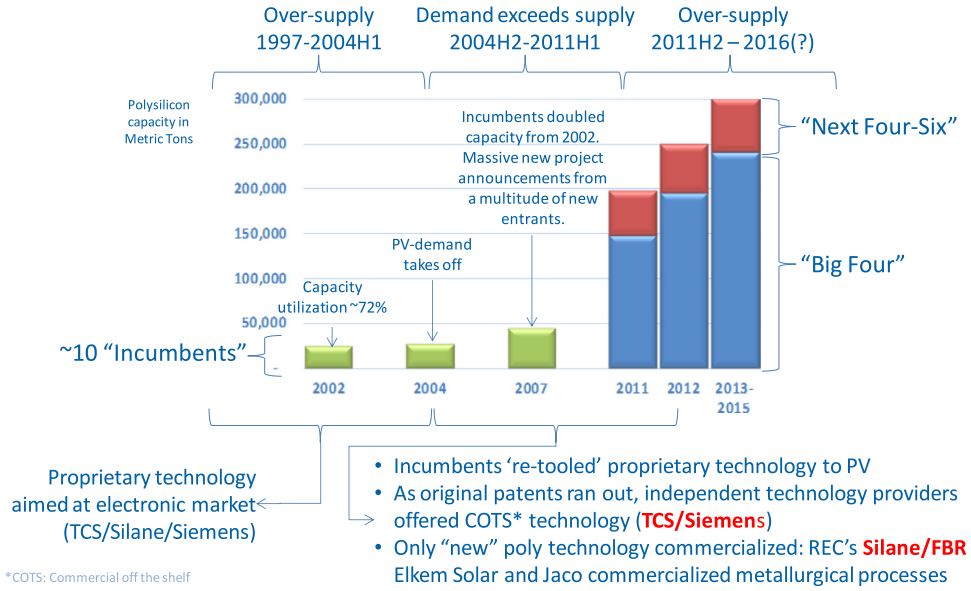


Fig. 5. Polysilicon industry in transition.

polysilicon facilities are part of integrated PV-concerns and thus mostly sell their products to companies within the same corporation. Thus, perception of costs will be very subjective for owners of existing capacity and we will continue to see plants operating even if the enjoyed sales prices do not seem to justify the cost of running them (to a disinterested outsider).

3. Industry in transition

3.1. Industry players change

Ten years ago the world's polysilicon production capacity was a little more than 25 000 MT/y and the industry was dominated by, and basically made up of, a small number of producers: Wacker Chemie, Hemlock Semiconductor, MEMC (now SunEdison), Tokuyama, Advanced Silicon Materials/ASiMI (now REC Silicon), Mitsubishi Polysilicon, Mitsubishi Materials, Sumitomo Titanium; all of these targeting electronic wafer customers and selling mostly only "off-spec" polysilicon to PV-wafer makers. All of these polysilicon producers had, and indeed still have, proprietary production processes and technologies developed over decades—MEMC: Silane/FBR; ASiMI: Silane/Siemens; the remaining: TCS/Siemens.

As the market shift became noticeable from around 2004, these legacy producers positioned themselves better to serve and exploit the high-growth PV-segment—admittedly at varying speeds and to varying degrees. The positioning consisted of "retooling" their proprietary technology and processes, removing bottlenecks in existing capacity and initiating new expansion projects. By 2007, retooling and de-bottlenecking had increased the incumbents' capacity and output to around 45 000 MT/y, while significant new capacity had yet to enter the market and the supply was not sufficient to feed the frenzied growth of PV-wafer production capacity.

2007–2008 saw massive new project announcements from a multitude of new entrants due to the perceived attractive industry economics that enabled funding to be found for this capital intensive chemical processing capacity.

Access to technology had, in the past, been a high barrier to entry for new entrants to the polysilicon market because the proprietary technology was not generally available for licensing. However, as original Intellectual Property protections expired and polysilicon sales prices went through the roof, independent technology providers began to offer polysilicon technology – Commercially off the Shelf/COTS – nearly without exception these offerings were TCS/Siemens. The legacy polysilicon manufacturers all continue to use their own proprietary technology, while the new entrants purchased not-differentiating, early generation technology from third-party equipment providers. In fairness, premier technology providers such as GT Advanced Technologies and Morimatsu (China) Group quickly developed very advanced TCS/Siemens CVD-reactors and provided such to particularly the large customers with multi-phase/multi-year expansion projects.

The only "new" polysilicon manufacturing technology to be successfully commercialized in the last decade was REC's silane/FBR, and "new" is in quotation marks because the general technology as such had been practiced by MEMC for 20 years at the time. Outside a strict definition of polysilicon, several companies introduced metallurgical processes to make solar grade silicon. To our best knowledge at the time of writing, two or three of these a.o. Elkem Solar (Norway) and Jaco (China) are in semi-commercial operation.

3.2. Increasing scale and concentration

From 2009 and onwards, truly new capacity entered the market to create a hefty oversupply from 2011 and onwards. When the delayed, but believable capacity under construction become operational in 2015, or at the latest by the end of 2016, the supply side will have grown more than 10-fold from 2002.

The cast of characters also has changed, but it should be noted that capacity concentration has increased with the scale of the most important producers in spite of the large number of attempted new entries into the market.

There are several benefits of being a large-scale producer. De-bottlenecking and incremental expansions typically yield significantly better Internal Rate of Return/IRR than initial capacity. The larger players have a considerable advantage by expanding their existing platforms in increments, each of which may be larger than a new green-field establishment. Competency, experience, basic operating systems etc. also favor larger producers.

As depicted in Fig. 5, there are currently four polysilicon producers that from a capacity point of view stand out from the rest—GCL Poly and OCI Company, both newcomers to the industry;

and the incumbents Wacker Polysilicon and Hemlock Semiconductor. On completion of their delayed and postponed expansion initiatives, these four will total around 270 000 MT/y of polysilicon capacity—electronic and PV-grade.

In the next tier there are four to six other producers each having less than half the capacity of each of the four largest—two are incumbents: REC Silicon and Tokuyama; the rest are newcomers like LDK Silicon, Daqo New Energy Corporation and TBEA Xinjiang Silicon. Not all of these produce at present and correspondingly there are other companies that do produce, but whose capacities are non-significant, and whose survivability as polysilicon producers may be questioned.

Still other projects are in the EPC-phase and, if successful, will challenge the second tier players. Some of the most interesting of these are backed by "large balance sheets" from other industries, e.g. Hanwha Chemical that currently is putting their first polysilicon plant – COTS TCS/Siemens – into operations, and Samsung's Joint Venture with SunEdison currently building their first polysilicon plant – proprietary silane/FBR. It is fair to expect that more large industrial concerns are looking to enter the industry and that new silicon majors may emerge before the attrition and consolidation has played out in the polysilicon industry.

3.3. Geographical change to China

With amazing speed and determination, China has developed into the dominant producer of PV-ingots, wafers, cells, and modules. Its share of global PV-wafer production is upwards of 90%. China is also rapidly emerging as a new major end-market for PV-modules, but China is still dependent on imported polysilicon.

In 2012, China imported more than 80 000 MT of polysilicon. In the first seven months of 2013, the corresponding number is more than 45 000 MT – even with substantial domestic polysilicon capacity, idled for months, being restarted in the second quarter – e.g. GCL, Daqo and ReneSola.

China has, over the last five years, given birth to four of the top 10 polysilicon players measured by production capacity. This was enabled by a deep pre-existing chemical processing and industrial value chain, and has in turn given rise to the further development of a polysilicon value chain including metallurgical grade silicon/MGS (more than 50% of global MGS capacity is Chinese), TCS, equipment, pipes and vessels, construction competency etc. As seen previously when Chinese companies broadly enter an industry, a cluster rapidly emerges to drive domestic growth and cost rationalization. The depth and strength of this domestic value chain must not be underestimated when discussing further geographical shift of polysilicon production capacity to China.

Some argue that China has no clear competitive advantage within polysilicon production because the traditional advantage from having low labor cost does not apply to a relatively low-labor intense industry like polysilicon. Further, the price of the most important input to polysilicon production – energy – is in general high in China. However, the cost of equipment and construction is generally much lower than in the West and capital expenditure thus becomes very competitive—this is where the positive effects of the domestic value chain come into play. While North-America can offer some of the lowest power- and natural-gas prices in the world, this is out-weighed by the price of MGS, equipment, bulk materials, and construction—we will later look at energy cost and capital expenditure versus the most relevant polysilicon technologies.

Due to political backing and the above mentioned "realities on the ground" Chinese polysilicon production will become dominant over the next few years, especially as it was recently mandated by the Chinese Ministry of Industry and Information Technology that expansion by introducing new and improved technology can be permitted and funded [3]. This focus on new and improved technology for the entire PV-value chain is to rectify the errors made in the over-investment spree of the last five years using non-differentiating and non-competitive technologies mostly in too small a scale (this is certainly the case for most of the newly minted polysilicon production plants).

4. Solar grade silicon technologies

4.1. TCS/Siemens: The dominant technology

The hitherto dominant technology consists of producing trichlorosilane/TCS from metallurgical grade silicon, purifying this molecule by several distillation and condensation steps and decomposing it in a thermal chemical vapor deposition/CVD reactor commonly called "Siemens reactor" with reference to the firm, which first in the 1960s developed this process for semiconductor needs (see Fig. 6). Approximately 90% of the globally installed polysilicon capacity and actual output is produced through this technology. This process to polysilicon generates significant amounts of by-products, essentially silicon tetrachloride/STC and other chlorosilanes, which need to be recycled in a closed loop process. Two process strategies are pursued to generate TCS and recycle STC:

- Low pressure (1–5 bars) and low temperature (300–350 °C) hydrochlorination of impure metallurgical grade silicon/MGS by hydrogen chloride for the synthesis of TCS and thermal (1300 °C) hydrogenation for recycling STC to TCS according to:
  - $Si\ (MGS) + 3HCl = SiHCl_3 + H_2$
  - $SiCl_4 + H_2 = SiHCl_3 + HCl$
- High pressure (20–35 bars) and high temperature (550 °C) hydrochlorination of MGS and STC in one step according to:
  - $Si\ (MGS) + 3SiCl_4 + H_2 = 4SiHCl_3$

The technical and economical competition between these two routes has been broadly debated during the 10 past years without concluding to the absolute advantage of one or the other. Both are still widely used in the industry including by the new entrants. The local and specific conditions to each project may decide the most attractive choice. In the following article we will not make further distinction between them.

High purity TCS is then vaporized and introduced in the deposition Siemens reactor. The gas is decomposed onto the surface of heated silicon seed rods, electrically heated to about 1100 °C building large rods of high purity silicon according to the main equilibrium equations:

- $2SiHCl_3 = SiH_2Cl_2 + SiCl_4$
- $SiH_2Cl_2 = Si + 2HCl$
- $SiHCl_3 + H_2 = Si + 3HCl$
- $SiHCl_3 + HCl = SiCl_4 + H_2$

The Siemens process is highly energy consuming, a major part of the energy being dispersed and lost. A major advancement in polysilicon production was the utilization of metal bell jars in place of quartz bell jars. It made it possible to accommodate 30 or more inverted U-seed rods in each reactor. This dramatically increased the productivity while decreasing the specific energy consumption per kg of produced polysilicon. Recent improvements of the Siemens process have focused on energy reduction and productivity increase by playing with the growth rate, the filament/rod configuration, the reactor size and -design. The number and the length of seed-rods have increased. The heat lost and wasted through the reactor wall is minimized by applying

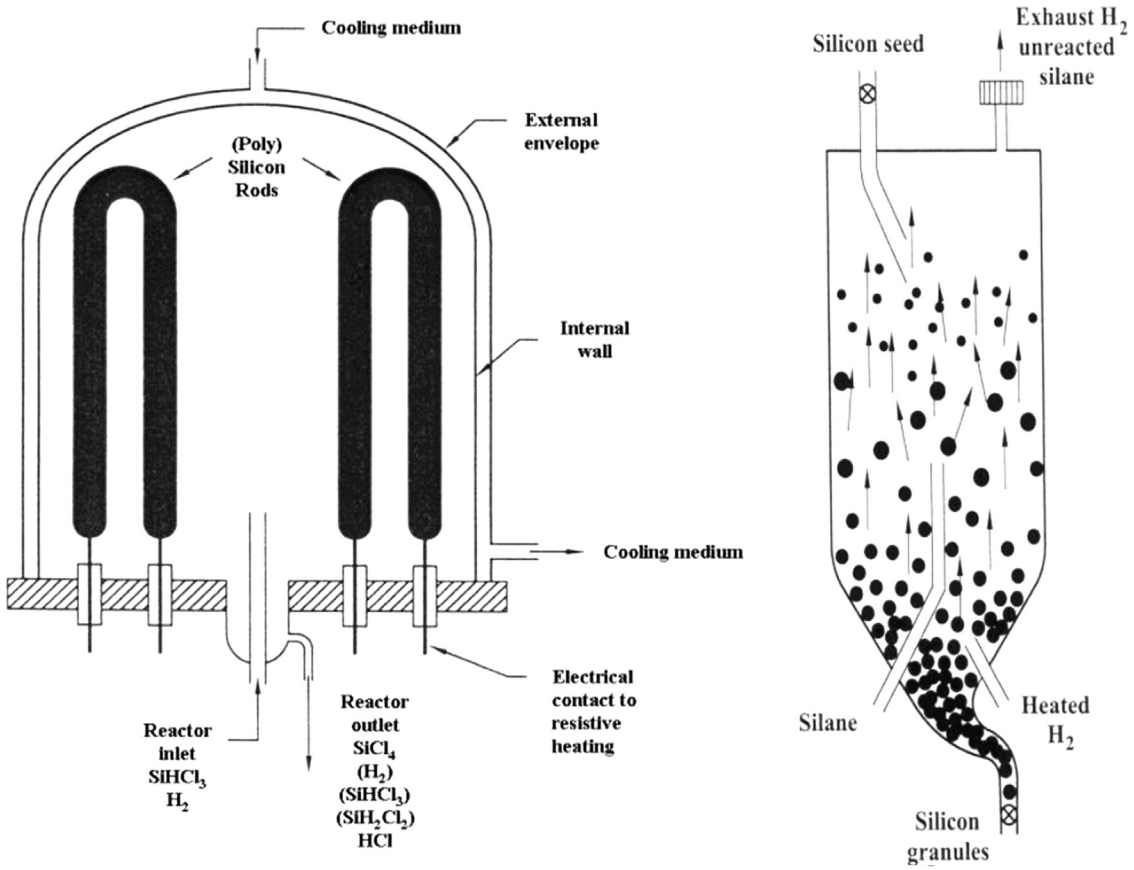


Fig. 6. TCS/Siemens the dominant polysilicon technology (left) and silane/FBR the challenging polysilicon technology (right).

thermal reflector on the inner wall of the reactor. These recent developments allow new production facilities to be built with Siemens reactors operating with 36–48 inverse U-rods or pairs of rods purchased COTS; the largest and most efficient reactors have an annual capacity of 450–600 t and consume below 50 kW h/kg polysilicon (see Section 6 on cost differentiation).

4.2. Silane/FBR: The challenger

Production of polysilicon in a fluidized bed reactor/FBR using monosilane/SiH_4/silane as feed-gas appears as a challenger to the above described TCS/Siemens process. The reason is the strong reduction of energy consumed in the deposition process, an order of magnitude lower than in the Siemens deposition. Less than 10% of the globally installed polysilicon capacity and actual output is generated through this technology. However, it is the subject of numerous development projects within both incumbent and new producers of polysilicon. FBRs are used in many industrial processes, for instance for gasoline production and coal gasification. Fluidized solid particles charged into a closed vessel may either be further grown or consumed, depending on the conditions and the reactants in presence. For a polysilicon process the first alternative is suited. Using silane as feed gas, 100% of it can be converted to elementary silicon with hydrogen gas as the only byproduct according to the equation:

- $SiH_4 \rightarrow Si + 2\ H_2$

"Seeds", in the form of fine particles, are (semi)continuously loaded from the top or the middle of the reactor, whereas silane gas and hydrogen are introduced near the bottom of the reactor (see Fig. 6). Ascending flowing gas percolates through the particle bed. When forces from the gas stream equal those of gravity, the particles begin to lift, making the bed behave as a fluid. The bed exhibits a large degree of temperature uniformity because the particles are in constant motion. Furthermore, the particles are moving freely, enabling a uniform chemical vapor deposition. Fluidized bed reactors are designed to work within a given regime determined by parameters like particle density and particle size distribution, bed height, gas flow and pressure. Because chemical vapor deposition is responsible for the growth process of the particles, temperature and surface area available for the deposition process become important parameters. As the particles grow, maintaining the fluidization becomes difficult. It is therefore necessary to remove the heaviest of the grown particles, pulled downwards in the bed by gravity. To keep the population, or the distribution, of particles in the bed in constant proportion and thus keep the bed in steady state condition, one needs to add a constant and continuous flow of seeds and likewise to remove the largest particles ("beads") to compensate the growth.

An obvious benefit of the FBR operational mode is continuous operations as both feed and exhaust gases on one hand and solid seed particles and finish granules on the other hand can be introduced and removed simultaneously. While typical batch/cycle time for Siemens reactors is 60–150 h, it is 60–120 days for FBRs.

FBR deposition of silicon presents some challenges. Formation of fines and powder becomes an issue when silane is used as feed-gas. Another challenge is the deposition of silicon on the wall of the reactor and other hot parts inside the reactor e.g. gas injectors etc. Both issues contribute to lower silicon yield, lead to unsuitable products, and product contamination. Modeling using Computational Fluid Dynamics/CFD and heat transfer calculations are important tools to understand chemical reactions and growth mechanisms occurring in the FBR. Results of such studies are suggesting new reactor design measures to overcome powder

generation and product contamination. Such measures include internal wall coating and alternative internal heating replacing, or supplementing, external heating through the wall. Internal heating can be achieved by means of microwave or through an internal reactor tube or also by favoring exothermic reactions and/or many other means. Which technical measures commercial companies have applied remain mostly undisclosed trade secrets as only a handful of companies are involved in this technology with its high entry barriers. From these companies' news releases and announcements we can deduct that large reactors of 1 000 t and more annual capacity are currently in operation and that specific energy consumption for CVD deposition is as low as 5–10 kW h/kg polysilicon, which is a good order of magnitude lower than the best-in-class Siemens reactor. Announcements have been made by the joint venture between SunEdison and Samsung in Korea to the effect that it is targeting deposition FBRs of several thousand tons (capacity per annum) each.

Both the current commercial producers of FBR polysilicon use silane as feed-gas. There are at least three commercial processes to make silane. The first one was developed by Union Carbide in the early 80s of last century and consists of the redistribution of purified trichlorosilane through fixed bed columns filled with quaternary ammonium ion exchange resins acting as catalyst:

- $2SiHCl_3 = SiH_2Cl_2 + SiCl_4$
- $3SiH_2Cl_2 = SiH_4 + 2SiHCl_3$

The reactants and products of the redistribution are separated by distillation. This first method is gaining most recognition for the production of silane regardless the purpose; polysilicon or other applications.

A second process, practiced by SunEdison (formerly MEMC and Ethyl Corporation in the USA) consists of the hydrogenation of silicon tetrafluoride SiF_4 by metal hydrides such as lithium-aluminum or sodium-aluminum hydrides:

- $2H_2 + M + Al = AlMH_4 \cdot M = Na, Li$
- $SiF_4 + AlMH_4 = SiH_4 + AlMF_4$

SiF_4 may be obtained by fluorination of natural silica or by sublimation of natural alkaline fluorosilicates (M_2SiF_6, M = alkaline element) which can be isolated from fertilizers' phosphates ores.

In a third method practiced in Japan by the Company Mitsui Toatsu a mix of silicon and magnesium powder is attacked by hydrogen chloride in presence of ammonia. Resulting products are silane and ammonium chloride salt:

- $Si + Mg + 3HCl + NH_3 + H_2 \rightarrow SiH_4 + MgCl_2 + NH_4Cl$

4.3. UMG: Emerging technology?

Stimulated by the shortage of high purity silicon feedstock in the period 2004–2008 numerous projects were launched to refine silicon up to an adequate purity for solar cells using the metallurgical routes. These initiatives were broadly inspired by similar attempts during the first wave of research and development on commercial low cost photovoltaics in the 70–80s of the last century. This class of processes and products are commonly designated in the industry as "UMG" standing for "Upgraded Metallurgical Grade (Silicon)". We use also this generic designation in the present article for any silicon purification process and product achieved by metallurgical routes.

Table 1 reports a short list among the most accomplished UMG projects. To our best knowledge, at the time of writing none of these projects is in regular continuous production. Two or three companies are pursuing a commercial activity a.o. Elkem Solar and Jaco, but the majority among these projects has been abandoned, the remaining being either in stand-by or pilot production mode. We estimate that the worldwide installed UMG capacity in 2012 was less than 15 000 MT/y (the most notable producer being Elkem Solar, now owned by the Chinese National Bluestar Group with a plant of 5–6000 MT/y in Norway) and the commercial output for that year in the order of 1000 MT—production output probably did not significantly exceed 5000 MT in total over the last four years.

Each company referred to in Table 1 uses specific proprietary technology. Each process is normally based on several steps each taking care of groups of impurities e.g. donor (P), acceptor (B), transition elements (Fe, Ni, Co, Cr, Ti, etc.), carbon (C) and less noble elements than silicon (Al, Ca). A majority of processes uses the following steps:

- Carbothermic reduction of quartz with careful selection of the raw materials to minimize the impurity content.
- Ladle refining using gas (alternatively plasma) or slag.

In spite of outstanding results achieved by the most advanced UMG products such as Elkem Solar Silicon (ESS®) the products under this category suffer from a high dopant level which limits their application as we will explain in the section Product Differentiation below. A serious consequence is that the usage of these products is so far recommended only in blend with higher grade polysilicon. However, these types of process are demonstrating astonishing progress and hold the promises of lower cost than Siemens and FBR (capital expenditure and lower energy consumption). Therefore, we decide for the purpose of this article to include this category with the question mark as the emerging technology.

4.4. Non confirmed technologies

Besides the three technology routes above (TCS/Siemens, silane/FBR, UMG) many other processes have been proposed and are still under investigation. Other volatile silicon bearing molecules than TCS and silane (silicon tetrachloride, silicon tetraiodide, tribromosilane, alkoxysilanes a.o.) and other reactor types (Free Space Reactor, Tubular Reactor, Vapor to Liquid Deposition Reactor a.o) are suggested. Without prejudging of the future success or failure of these processes we will not further discuss them in the present article as none of them has reached a satisfactory industrial maturity.

For a further study of all solar silicon feedstock processes the interested reader is invited to consult more comprehensive articles [3–5].

4.5. Technology availability

Before 2002 all polysilicon and UMG technologies were well protected by the legacy producers of polysilicon for the former and the most capable MGS producers for the latter. With the rapid growth of the photovoltaic market and the subsequent shortage of silicon feedstock in the period 2004–2008 the TCS/Siemens technology was rapidly disseminated, particularly among new entrants in Asia (China and Korea). Although not so widely disseminated, silane technology is currently also accessible from several technology providers and chemical engineering firms. High investment costs and long planning, construction and ramping periods (3 to 4 years) are limiting the access to this technology. However, it is worth remembering the speed with which particularly Korean and Chinese new entrants ramped production using TCS/Siemens.

FBR and UMG are not available from any vendors as per now. Developing an FBR or a UMG process nominally takes 5 to 10 years of development and millions of dollars without guaranteed success.

5. Differentiation by product

High purity silicon produced by the above described technologies are however not all equivalent. They can be broadly differentiated both by their morphology and purity.

5.1. Morphology

Siemens polysilicon is generated as cylindrical rods (see Fig. 7) and rod sections 140–150 mm in diameter and up to 2.5 m in length. For practical purposes the rods and rod sections need to be further crushed to larger and smaller lumps of 10 to 100 mm commonly called “chunks” and 1 to 10 mm commonly called “chips” (rod-pieces measuring 200–400 mm can also be used “as-is”).

FBR polysilicon is generated as small spheroid particles ranging from 0.1 mm to 1 or 2 mm in size (see Fig. 8). Depending on the reactor design and process parameters chosen the average size is around 0.5 to 1 mm. As UMG categorizes many different processes, the products may have different shapes and aspects. If the last steps are melting followed by casting an ingot is obtained. For practical purposes this ingot needs to be crushed into lumps and chips or sawn into bricks and plates.

Table 1
Overview of the most accomplished UMG silicon projects (2000–2012).

Company (Country)	Status
JFE (Japan)	Stopped due to high cost. May be reopened if needed
Dow Corning (Brazil)	Stopped
Timminco (Canada)	Stopped. Technology sold to Ferroatlantica
Elkem Solar (Norway)	Production temporarily halted. Development on going
Jaco (China PR)	Production at pilot scale level
Silicor Materials (Canada, USA)	Stopped. May be restarted
Photosil (France)	Pilot
Ferroatlantica	Pilot
Evonik/Solsic (Norway)	Stopped irreversibly

5.2. Purity

The main differentiation seems however to be on purity. The metallurgical processes have not proven yet as powerful as chemical distillation and condensation to eliminate the donors and acceptors (essentially phosphorous and boron). Table 2 gives an overview on the typical purity level by products and types of impurity.

In UMG contents of donors and acceptors are 2 or 3 orders of magnitude higher than in polysilicon refined by gas distillation/condensation, whereas metal impurities can be as low in UMG as in polysilicon. The high donor/acceptor concentrations impose severe restrictions and constraints on how to use UMG as feedstock to solar cells (see Section 5.3).

FBR allows low dopant concentrations as purification goes through gas phase. However, FBR is less efficient than the Siemens reactor with respect to metal, carbon and oxygen contamination [6–7]. In FBR metal contamination may be caused by wall deposits, friction and contact with the vessels, pipes and valves which are inevitable to enable that process to take advantage of the fluid nature of the material. Seed generation through milling is also a source of contamination. Finely divided materials offering a large specific area like granular polysilicon are naturally more oxidized and the surface tends to attract more ambient impurities like



Fig. 8. FBR polysilicon granules.



Fig. 7. Polysilicon rods as produced by TCS/Siemens.

Table 2
Differentiation of silicon products by purity.

Impurity	Siemens (solar) (value range)	FBR (value range)	UMG (value range)
P (donor)	0.3–5 ppba	0.3–20 ppba	300–1000 ppba
B (acceptor)	0.1–5 ppba	0.3–20 ppba	500–2000 ppba
Total metals	20–50 ppbw	30–1000 ppbw	10–1000 ppbw
C	0.25–1 ppma	0.5–10 ppma	50–200 ppma
O	0.5–5 ppmw	10–100 ppmw	(100 ppmw)
Gas inclusion		H_2	

carbon than bulk polysilicon does. As we will see in the following, shape, size and purity differences will introduce differentiations in using these products when manufacturing solar wafers and cells.

5.3. Domains of applications

High purity silicon is for the manufacture of solar cells further processed into ingot and wafers. The dominant technologies to make ingots are both the single crystal Czochralski/CZ technique and the multicrystalline/m-C directional solidification/DS. CZ is particularly suitable for high efficiency cells as these require a lower content of impurities and crystal defects whereas m-C is satisfactory for most of the standard solar cells. CZ made wafers can be either p or n-types whereas m-C made wafers are so far solely p-type. Each of these crystallization or pulling techniques is the subject of interesting recent developments. To increase productivity and reduce cost of manufacture, longer ingots or longer runs are achieved by adding more silicon to the crucible before and during the melt phase. Semi-continuous CZ is already current practice and continuous CZ is under development. Using DS machines "quasimono" and "monocast" crystals are achieved under certain conditions leading to superior quality for the resulting multi-crystalline wafers, thus closing the gap between the performances of CZ and m-C wafers when converted to solar cells. These developments in ingot, wafer and cell technologies are strongly dependent on the feedstock quality and attributes.

Charging a Bridgman crucible for m-C with silicon lumps from the standard Siemens process takes a skilled workforce to avoid scratching the surface coating. If silicon melt comes in direct contact with the silica crucible the melt may be contaminated and the ingot may crack during cooling due to sticking. Large rod sections will be preferred as the same time as they optimize the filling factor. Using granular feedstock as obtained by FBR avoids such problems. Furthermore, granular material allows more automation in handling (less labor), and facilitates maximization of the crucible load by refilling and/or filling the voids between the lumps. For specific applications granular material is particularly suitable e.g. continuous feeding and refill. Reports from use of granular silicon in the market place indicate that initial crucible loads should not contain more that around 40–50% granular material due to impurities and heat transfer in the melt.

Because of the high level of donors and acceptors, UMG is a "compensated" material indicating that most of the charge carriers neutralize each other, but the high level of the respective dopants disturbs the semiconductor properties by creating molecular interactions whose effects are not fully understood. Another inconvenience expected with 100% UMG as feedstock, is lower material yield during ingot production driving up the cost of the ingot. This is due to a significant part of the ingot having the wrong concentration of donor due to p- to- n-type conversion in the brick. This can be corrected and fine-tuned with the use of gallium as a co-dopant. It is possible to control the resistivity and conductivity type in the upper part of the ingot by the use of gallium as co-dopant, so the lower material yield issue is technically solvable, as shown in several papers [8–9]. However, UMG is mostly recommended when it can be blended with conventional polysilicon in a ratio varying from 20 to 80% although 100% have been tested with good result for the user. Another negative impact of using compensated material with high boron level is the risk of light induced degradation (LID) caused by the B–O molecular pair. This has been pointed out in several papers although the fine mechanism of degradation is not fully understood and still debated [10–11]. Until now UMG is used only in p-type m-C wafers for standard cells.. Results from the fields show that anticipated drawbacks such as light induced/LID and other degradation effects are not consistently demonstrated in these fields. Under high real climatic irradiation condition Elkem Solar claims that cells made with its UMG were superior in producing amount of electrical current [12].

All these developments indicate room for different sorts of feedstock especially if this contributes to an overall decrease of the cost as the next section will explain.

6. Cost of the technology options

Cost is an area where one has to tread lightly and be careful to ensure analyses compare "apples to apples".

There is no one best way to define "cost", or indeed one generally accepted way of relating to "cost" when discussing technology options and price and cost developments. This is why comparisons of production plants, technologies, and processes only relying on the various statements of "cost per kg" from not-disinterested parties are fraught with inaccuracies and errors.

When capacity utilization was high, fixed capacity costs were diluted by the high production volumes, and producers and technology providers took much pride in reporting the low levels of cost per kg—not so much when capacity is underutilized and fixed costs become more significant. The most prominent fixed cost is typically the depreciation (over the assumed lifetime of a production facility) of the capital expenditure for building said production capacity.

In recent time, when production capacity has been underutilized and the carried assets value on company balance sheets has been written down or completely written off, the focus has mostly turned to "cash cost". Cash cost is a term meaning the actual expenses one has to pay while operating a production plant, in contrast to the calculated depreciation cost that already has been expensed from the gestation to the potential subsequent debottlenecking of the operating production capacity. Cash cost will thus give a picture of the cost of continued operation and also at which minimum sales price it is sensible to operate a particular production plant: if the sales revenue covers cash cost, the cash flow to the plant owner is positive.

It is highly subjective at which level a producer chooses to measure cash cost, particularly when it relates to an "operate/do not operate" decision on existing production capacity. An idle production facility also incurs cash cost e.g. if the workforce is retained. Dependent on the producer's view of the future demand and price developments, it may make sense to operate a production facility even if the sales price only covers raw materials and utilities.

As we need to compare the cost of the various technology options for future capacity expansions in light of the developments in the industry, we nevertheless boldly go forth and take a closer look at the two largest cost drivers for polysilicon production: Energy consumption and capital expenditure. We also include the most advanced UMG technology for the reasons discussed when presenting this technology earlier in this article.

6.1. Energy consumption

In spite of the confusion on absolute cost, there is general agreement that the single largest operational cost driver for solar grade silicon production is energy consumption. Large energy consumption impacts negatively silicon economics, energy payback time and carbon emissions of PV. Discussions often focus on power consumption only, but this would omit the significant need for thermal energy that is delivered by burning natural gas, diesel, or coal. In many geographies the access to, and the pollution from, power generation and feedstock for thermal energy is unsustainable as well as costly.

The ease with which energy can be obtained as well as the effective price paid for it vary from locale to locale and will determine how important energy consumption will be perceived when prospective producers evaluate technologies. However, in general a lower energy consumption is to be desired.

Fig. 9 shows typical energy consumption for the two main polysilicon technologies and for the most advanced UMG-technology. For the polysilicon processes the energy consumption is broken down into power usage for the CVD-reactor and power and thermal energy consumption for the remainder of a production plant, i.e. feed-gas generation, gas recapture and recycle, utilities etc. For UMG processes no feed-gas or CVD are utilized. For all technologies the total energy consumption covers all sub-processes from introducing MGS-raw material in the process to finished solar grade product ready for the crucible.

A state of the art the TCS/Siemens process consumes a total of about 130 kW h per kg polysilicon produced. The CVD-reactor sub-process consumes around 45 kW h [13] and the balance of the plant around 85 kW h in a combination of electricity and thermal energy [14]. By utilizing heat reflector coating and heat recapture in the CVD process to offset thermal energy usage elsewhere in the process, it is (at least in theory) possible to reduce the total energy usage down to 100 kW h per kg polysilicon produced—as indicated by the black line inside the column for TCS/Siemens in Fig. 9.

Silane/FBR consumes much less power in the CVD-sub process due to the lower deposition temperature of silane and no cooling of the reactor wall—around 5 kW h by kg polysilicon produced is reported [15]. Also the balance of a silane plant consumes less energy, particularly electricity [16], than a TCS process because there are less chemical constituents to recapture and recycle from the CVD exhaust gas – the only byproduct is hydrogen – and thus fewer sub-processes. Thermal energy usage is on the other hand on par with the TCS-process [17]. Total energy consumption closes in on 55 kW h per kg polysilicon produced. The flipside of the lower deposition temperature in the FBR is less possibility to recapture and reintroduce heat.

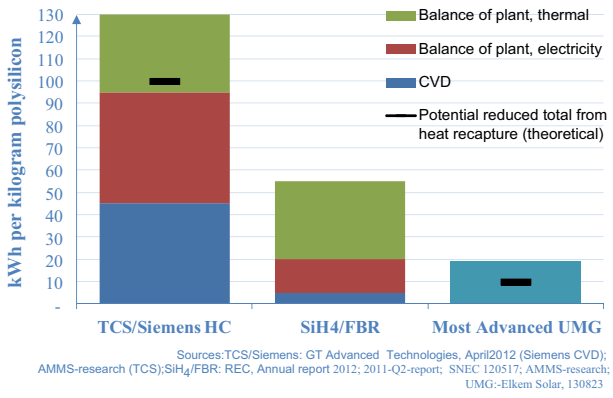


Fig. 9. Energy usage for TCS/Siemens, silane/FBR and UMG.

The most advanced UMG-process reports the lowest energy usage at a total of 20 kW h per kg solar grade silicon produced [18]. The process owner claims to have a next generation technology under development eliminating the expensive and energy consuming purification through directional solidification. If successful, this should cut the energy consumption in half—see the indicative black line in the UMG-column.

6.2. Capital expenditure

For new capacity, it is glaringly obvious that the investment decisions must take into account the total cost of ownership over the lifespan of the production plants and not only the "cash cost". The level of initial capital expenditure will impact not only the absolute amount of investment tied up in the production assets; there are important relationships between the cost of constructing a polysilicon plant and future ongoing cost. Initial capital expenditure drives the level of a.o. finance cost and maintenance expense. Unlike the calculated depreciation cost, these related and payable costs do not go away even if the assets should be written down due to financial hardship.

In general, a lower initial capital expenditure is therefore to be desired.

6.3. Technologies compared on cost

In Fig. 10 we have depicted the cost status and potentials for the technologies we have discussed above.

Along the *x*-axis is a combined cash cost where cautiously calculated maintenance expense and finance cost have been added to the variable cost of production as reported by various sources. Ongoing maintenance expense is calculated as 4% of initial capital expenditure. Finance cost is calculated as 2% of total initial capital expenditure.

Along the *y*-axis is the initial capital expenditure in USD per kg of annual production capacity. Straight line depreciation over 10 years is used to derive depreciation cost per kg of production over the lifespan of the production facility.

Each of the diagonal lines represent combinations of capital expenditure – turned into depreciation – and cash costs that total the same amount – from left to right the lines represent respectively USD 12.50 per kg, USD 17.50 per kg, USD 22.50 per kg and USD 27.50 per kg.

The solid symbols represent the best in class cost for existing and de-bottlenecked production capacity using each of the three technologies. The corresponding outlined symbols represent the cost potential for the same three technologies assuming that green-field capacity would be built in China today.

For all the technologies, existing best in class production capacity has a total cost of around USD 27.50 per kg polysilicon produced. As can be seen, the dominant and most mature technology has lower capital expenditure than the two others for which the term "best in class" capacity is somewhat misleading in this regard because the actual production plants used to report capital expenditure in fact both are one-of-a-kind production facilities built with never before used new technology, and not necessarily representative of the cost of constructing subsequent facilities with the same technology.

Theorizing over the construction of new production facilities using further improved versions of the three technologies as of today, a more differentiated picture emerges. Indications are that the TCS/Siemens technology is nearing its limits for cost improvement, while the two other technologies still have significant improvement potential.

An amalgamation of various announcements and proprietary project estimates indicates that it would cost around USD 55 per

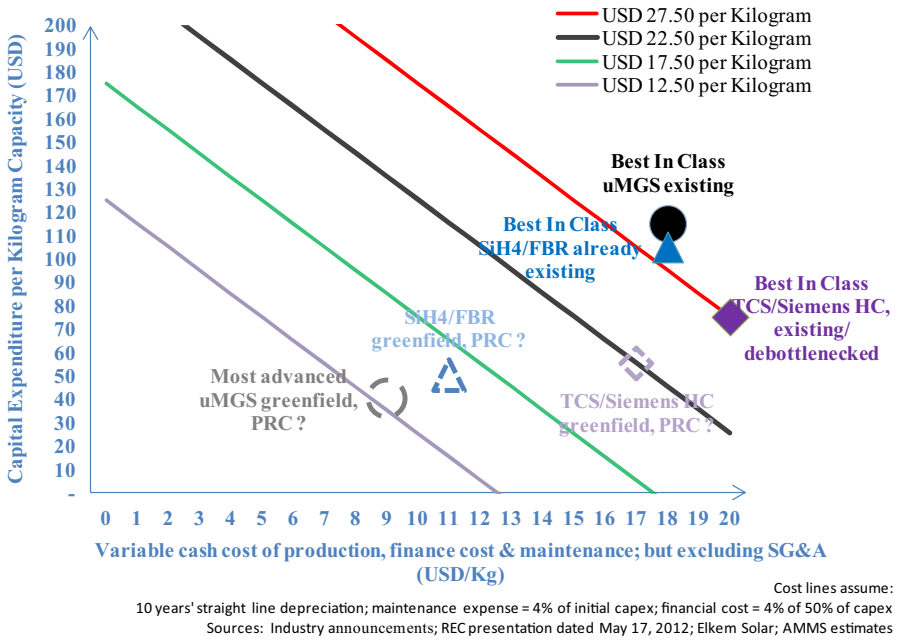


Fig. 10. Cost comparisons TCS/Siemens, silane/FBR and UMG.

Table 3
Solar grade silicon technologies side-by-side.

Sources: company announcements	TCS/Siemens	Silane/FBR	UMG (most advanced)
CAPEX new capacity	GCL, OCI: **35**[a]**–70 USD/kg**	REC (2006–2010): **105–120**[f] **USD/kg**	(2006–2009): **110–120 USD/kg**
	TKY, WCH, HSC: **90–120 USD/kg** China, green field: < **60**[b] **USD/kg**	China, new plant: **50**[g] **USD/kg**	China, new plant existing technology: **50 USD/kg** next generation: **33 USD/kg**
Cash cost (x/SG&A, maintenance and finance cost)	**18–23**[c] **USD/kg**	REC (2013 Q4e): **11.5 USD/kg**	**11 USD/kg**
Power usage (best, CVD only)	**35–45**[d] **kW h/kg**	REC: **5 kW h/kg**	**N/A**
Quality	**Solar** **Electronic** **"Real" electronic**	REC: **Solar**[h] SUNE: **"Real" electronic**	**Solar (w/compensated donor/acceptor levels)**
Installed base WW	**275–300 kMT/y**	~**25 kMT/y**	~**15 kMT/y**
Largest single site	**45–65** [e]**kMT/y**	~**20** [e]**kMT/y**	~**6 kMT/y**
Largest single CVD-rx	> **500 MT/y**	> ~**900** [e]**MT/y**	**N/A**
Years commercial application	> **50 years**	SUNE: ~**25 years** REC: ~**4 years**	~**4 years**
Availability	**Proprietary:** incumbents	**Proprietary:** SUNE/REC (Others: not quite yet-GCL, Samsung, SET)	**Proprietary:** Elkem Solar, JACO
	COTS: GTAT, Shanghai Morimatsu	**COTS:** not quite yet	**COTS:** no

[a] De-bottlenecking/incremental additions to existing facilities.
[b] Dependent on quality ambitions.
[c] Dependent on actual power price.
[d] Somewhat dependent on reflective bell jar coating, how heat recovery is accounted for etc.
[e] GCL Poly/REC silicon III; 2013.
[f] REC '11 Q4: Depreciation 2011–2012=USD 8–7/kg; 15 year straight line.
[g] REC presentation to SNEC—May 17, 2012.
[h] Solar is the major polysilicon segment.

kg of annual capacity to construct a new green-field TCS/Siemens plant of around 30 000 MT/y capacity in China. Utilizing all available energy conservation and other improvements, the cash cost per kg should be able to reach USD 17 per kg leading to total cost per kg of USD 22.50 down from existing capacity's USD 27.50.

According to REC Silicon a 20 000 MT/y plant based on their newest generation silane/FBR-technology could be constructed in China for around USD 50 per kg of annual capacity. Adjusting their estimate of cash cost with the effect of finance cost and maintenance expense as explained above leads to a total cost of around USD 16 per kg of polysilicon produced.

Finally, Elkem Solar indicates that their next generation technology could be constructed in China for around USD 40 per kg of annual capacity assuming 15 000 MT/y. Adjusting the projected cash cost as explained above would lead to a total cost as low as USD 13 per kg of UMG produced.

7. Technology comparison: Summary

Table 3 summarizes and compares side-by-side the various aspects of the technologies discussed in this article.

8. Concluding remarks

We assume that crystalline silicon technologies for the foreseeable future will remain dominant in the manufacture of solar cells. In this article we have shown that solar grade silicon should be available at an affordable cost to allow the photovoltaic industry to respond to the global energy challenge. By "affordable" cost we mean that a price of 20 USD per kg solar grade silicon will allow the industry to target an average sales price for PV-modules of 0.45 USD/Wp by 2020—as indicated by extrapolating the historic price experience curve for PV-modules.

Over-capacity puts pressure on the polysilicon prices to a recent and present level well below the above longer term target price. We are expecting balance recovery between polysilicon demand and supply by 2017, with periodic supply tightness already from 2014 due to not-perfect timing of the suppliers' bringing currently idled capacity and delayed expansion projects into operation. Therefore, we suggest that new capacity should be added from 2017 and onwards. More important is that a significant portion of the installed capacity does not seem financially sustainable at a price of 20 USD per kg. Not competitive capacity should be phased out and replaced by more competitive plants and technologies. In total, these conclusions indicate significant investments in solar silicon capacity over the next few years. We foresee continued increases in scale and concentration as well as a continued geographical shift to China.

We came to the conclusion that the state-of-the-art trichlorosilane based Siemens deposition is nearing its limits for cost improvement and is facing competition from silane based fluidized bed/FBR and metallurgical refining/UMG as these latter technologies consume far less power and necessitate less capital. However, we want to remain careful in our predictions which might be largely influenced by factors not thoroughly investigated in this article. Taking the above into account, one has to wonder:

- For how long will crystalline silicon remain the dominant technology for solar cells? Will specific consumption of silicon reach, and then possibly stagnate at, 4.5 g/Wp?
- Can TCS/Siemens reach new cost limits and thus justify building new capacity based on this technology?
- Can silane/FBR deliver on the cost and quality promises, particularly for technology from emerging third party technology providers? May silane/FBR granules eventually be used 100% in high performance PV-products?
- Will the upgraded metallurgical silicon process overcome the impurity stigma and will it ever become widely commercially available? Will higher performance cells be feasible with UMG?

For those actively participating in the polysilicon industry over the last 10 years, it must have been a fascinating journey. The industry developed from being an arena for a few especially interested participants; through an exponential growth and the entry of a multitude of new hopeful participants using traditional technology. We believe the industry is now on the brink of a "third wave": The next 10 years will be characterized by continued strong volume growth (again concentrated to a limited number of dominant players), profit margins more in line with commodity industries (that polysilicon strongly resembles), and with true innovation and technology changes. We predict that the third wave will be equally fascinating as the former 10 years.

References

[1] Citi, Global Perspectives & Solutions: Energy Darwinism, October 2013.
[2] BNEF/Bloomberg New Energy Finance, Solar Research Note, February 2013.
[3] MIIT/Chinese Ministry of Industry and Information Technology, September 2013.
[4] B. Ceccaroli, O. Lohne, Solar grade silicon feedstock, in: Antonio Luque, Steven Hegedus (Eds.), Handbook of Photovoltaic Science and Engineering, first ed.,Wiley, 2003, pp. 153–204 (Chapter 5).
[5] B. Ceccaroli, S. Pizzini, Processes, in: Sergio Pizzini (Ed.), Advanced Silicon Materials for Photovoltaic Applications, Wiley, 2012, pp. 21–78 (Chapter 1).
[6] B. Ceccaroli, O. Lohne, E.J. Øvrelid, New advances in polysilicon processes correlating feedstock properties and good crystal and wafer performances, Phys. Status Solidi C 9 (2012) 2062–2070.
[7] J. Li, G. Chen, P. Zhang, W. Wang, J. Duan, Technical challenges and progress in fluidized bed chemical vapor deposition of polysilicon, Chin. J. Chem. Eng. 19 (5) (2011) 747–753.
[8] M. Forster, E. Fourmond, R. Einhaus, H. Lauvray, J. Kraiem, M. Lemiti, Ga co-doping in Cz-grown silicon ingots to overcome limitations of B and P compensated silicon feedstock for PV applications, Physica Status Solidi C 8 (3) (2011) 678–681.
[9] A. Cuevas, M. Forster, F. Rougieux, D. Macdonald, Compensation engineering for silicon solar cells, Energy Procedia 15 (2012) 67–77.
[10] R. Kopecek, J. Arumughan, K. Peter, E.A. Good, J. Libal, M. Acciarri, S. Binetti, in: Proceedings of the 23rd European Photovoltaic Solar Energy Conference, Valencia, Spain, 2008. p. 1855.
[11] D. MacDonald, F. Rougieux, A. Cuevas, B. Lim, J. Schmidt, M. Di Sabatino, L. J. Geerligs, J. Appl. Phys. 105 (2009).
[12] Odden J.O., Lommason T.C. Tayyib M., Buseth T., Vedde J., Friestad K., Date H., Trondstad R., Results on performances of solar modules based on Elkem Solar Silicon from installations at various locations, in: Communication at Second Silicon Material Workshop, 7th and 8th October, Rome, 2013.
[13] GT Advanced Technologies, in: D. Keck (Ed.), Lowering the Cost to Produce Polycrystalline Silicon, Presentation to Photon Silicon Conference, 26th March, Berlin, 2012.
[14] Amalgamation of Empiric Operational Data Gathered by AMMS/Authors from Various Consultancy Projects for Polycrystalline Silicon Producers—Each Data Set is Proprietary and Confidential Business Information for the Various Polycrystalline Producers.
[15] R.E.C., Torvund T, Granular Polysilicon Technology, Presentation to SNEC, 17th May, Shanghai, 2012.
[16] REC, Second quarter presentation, 19th July, Oslo, 2011.
[17] REC, Annual Report 2012 (Derived from Sustainability Report and production numbers).
[18] Elkem Solar, Ulset T, Private & Confidential Communication to the Authors, 23rd August, Kristiansand, 2013.